

Education

Olin College of Engineering (B.S. Electrical and Computer Engineering) • Needham, MA • GPA: 4.0	<i>May 2029</i>
Relevant Coursework: Linear Algebra, Multivariable Calculus, Modeling and Simulation, Software Design	

Experience

Olin AERO (Autonomous Flight Team) • High Voltage Lead • Needham, MA	<i>Aug 2025 – Present</i>
- Building the school's first autonomous VTOL drone for 2026 Student Unmanned Aerial Systems Competition - Designing and building an STM32-based Battery Management System (BMS) to monitor and control the performance of a 12s4p Li-ion battery pack with galvanic isolation and overcharge/discharge, overcurrent, and short-circuit protection - Designed custom PCB schematics and layouts in KiCAD for electrical systems in drones - Troubleshooted electrical outputs using a signal generator and oscilloscope	
80V Analog Buck-Boost DC/DC Controller • Needham, MA	<i>Nov 2025 – Present</i>
- Recreating a fully analog implementation of the LT8705 from discrete components with a student team of four - Designed a voltage-controlled oscillator in LTspice using transconductance amplifiers for clocking the system - Created high-speed, high-side and low-side NMOS gate drivers using analog edge detectors and totem pole drivers	
Olin Rocketry • Principal Avionics Engineer • Needham, MA	<i>Aug 2025 – Present</i>
- Used an ESP32 microcontroller to perform sensor functional testing and diagnostics - Executed layup of carbon fiber composite materials for a lightweight, high-strength rocket body	
Wikipedia Speedrun Project • Linear Algebra Course • Needham, MA	<i>Nov 2025 – Dec 2025</i>
- Developed Python script to parse SQL dump files, cleaned and processed resulting data using Pandas - Modeled Wikipedia as a directed graph in MATLAB in order to analyze article connectivity and structure - Applied graph theory to determine shortest paths and walk counts, identifying “central” articles for more efficient speedruns - Computed resistance distances with in/out-degree Laplacians to quantify the difficulty of navigating between articles	
CAMP for Algorithmic and Mathematical Play • Counselor & TA • Albuquerque, NM	<i>Jun 2024, Aug 2025</i>
- Presented a thirty minute lecture on game theory, level-k reasoning, and the Nash equilibrium to 80+ campers and parents - Created and supervised activities for 15+ campers on pi approximation and games with math	
American Computer Science League Team • Co-founder • Reno, NV	<i>Aug 2022 – Aug 2024</i>
- Taught Python data structures and prepared PowerPoint slides for weekly team meetings - Placed as the #1 Nevada team with outstanding achievement in the American Computer Science League (2023)	
Carpenter Park Recreation Center • Lifeguard • Plano, TX	<i>May 2022 – Aug 2025</i>
- Saved two young swimmers from drowning and vigilantly supervised swimmers of all ages at the city pool - American Red Cross Certified Lifeguard Instructor / Lifeguard / First Aid / CPR / AED	

Skills

Programming & Development:	Python, C++, JavaScript, Git, MATLAB
PCB Design:	KiCAD, LTspice, Soldering, Oscilloscope, Logic Analyzer, Signal Generator
Microcontrollers:	Raspberry Pi, Arduino, ESP32, STM32, Linux
Tools:	Manual Mill/Lathe, 3D Printer, Band Saw, Drill Press